

41107 Marine and Ocean Engineering

Waves / Wave Theory / Wave Statistics / Spectral Analysis

Solved MCQ Question Bank - professional formula notation

Scope and ordering

This PDF collects the reviewed questions belonging to the **Waves** block, including **wave theory**, **wave statistics**, **spectral moments** and **spectral analysis**. Exact duplicates across the reviewed MCQ documents were removed. Same pattern with different numerical values is kept as a separate variant. The questions are ordered as follows:

1. Wave theory and regular-wave kinematics.
2. Wave statistics and extreme values.
3. Spectral moments and spectral response.

For each question, the structure is:

1. Original question statement and original answer options.
2. Correct answer highlighted immediately after the options.
3. **Formulario / principles used** with professional mathematical notation.
4. **Step-by-step solution** with numerical substitution whenever numerical data are present.

Default constants and notation

$$g = 9.81 \text{ m/s}^2, \quad \rho = 1025 \text{ kg/m}^3, \quad 1 \text{ knot} = 0.51444 \text{ m/s}$$

$$\omega = \frac{2\pi}{T} = 2\pi f, \quad f = \frac{1}{T} = \frac{\omega}{2\pi}, \quad k = \frac{2\pi}{L}$$

$$\text{Deep water: } L_0 = \frac{gT^2}{2\pi}, \quad c = \frac{L}{T} = \frac{\omega}{k}$$

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Wave theory and regular-wave kinematics

W1. Deep versus shallow water

WAVE THEORY - Deep vs Shallow Water. For deep-water waves, which of the following best describes the relationship between wavelength L and water depth h ?

Options:

- A. $h > L/20$ (deep water: $\tanh(kh) \approx 1$)
- B. $h < L/20$ (shallow water: $\tanh(kh) \approx kh$)
- C. $h = L/2$ (intermediate depth)

Correct answer: A. $h > L/20$ (as labelled in the source question)

Important correction. The physically standard criterion is $h > L/2$ for deep water and $h < L/20$ for shallow water. The source option A is internally inconsistent because it writes $h > L/20$ but labels it as deep water with $\tanh(kh) \approx 1$. For exam solving, remember the physical rule: $h > L/2 \Rightarrow$ deep water.

Formulario / principles used

- General dispersion relation:

$$\omega^2 = gk \tanh(kh)$$

- Wave number:

$$k = \frac{2\pi}{L}$$

- Deep-water limit:

$$h > \frac{L}{2} \Rightarrow kh > \pi, \quad \tanh(kh) \approx 1$$

- Shallow-water limit:

$$h < \frac{L}{20} \Rightarrow kh < \frac{\pi}{10}, \quad \tanh(kh) \approx kh$$

Step-by-step solution

1. Deep water means that the seabed has almost no influence on wave motion.
2. Mathematically, that corresponds to:

$$\tanh(kh) \approx 1.$$

3. Since $k = 2\pi/L$, deep-water behaviour requires:

$$kh = \frac{2\pi h}{L} \gtrsim \pi.$$

4. Therefore:

$$\frac{2\pi h}{L} \gtrsim \pi \Rightarrow h \gtrsim \frac{L}{2}.$$

5. Among the given options, the source labels option A as the intended deep-water option, but the correct physical criterion is $h > L/2$.

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

W2. Deep-water phase speed and group velocity

WAVE THEORY - Deep Water Wave Speed and Group Velocity. A regular deep-water wave has period $T = 10$ s. What is the phase speed c and the group velocity c_g ?

Options:

- A. $c = 15.6$ m/s, $c_g = 7.8$ m/s
- B. $c = 7.8$ m/s, $c_g = 15.6$ m/s
- C. $c = 15.6$ m/s, $c_g = 15.6$ m/s

Correct answer: A. $c = 15.6$ m/s, $c_g = 7.8$ m/s

Formulario / principles used

- Deep-water wavelength:

$$L_0 = \frac{gT^2}{2\pi}$$

- Phase speed:

$$c = \frac{L_0}{T} = \frac{gT}{2\pi}$$

- Deep-water group velocity:

$$c_g = \frac{c}{2}$$

Step-by-step solution

1. Calculate phase speed:

$$c = \frac{gT}{2\pi} = \frac{9.81 \cdot 10}{2\pi} = \frac{98.1}{6.283} = 15.6 \text{ m/s.}$$

2. For deep-water waves, the group velocity is half the phase speed:

$$c_g = \frac{c}{2} = \frac{15.6}{2} = 7.8 \text{ m/s.}$$

3. Therefore the correct pair is:

$$c = 15.6 \text{ m/s,} \quad c_g = 7.8 \text{ m/s.}$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

W3. Maximum horizontal particle velocity at the surface

WAVE THEORY - Particle Velocity and Acceleration in Regular Waves. A regular deep-water wave has height $H = 2$ m, period $T = 8$ s. What is the maximum horizontal particle velocity U_m at the surface ($z = 0$)?

Options:

- A. $U_m = 0.39$ m/s
- B. $U_m = 0.785$ m/s
- C. $U_m = 1.57$ m/s

Correct answer: B. $U_m = 0.785$ m/s

Formulario / principles used

- Wave amplitude:

$$a = \frac{H}{2}$$

- Angular frequency:

$$\omega = \frac{2\pi}{T}$$

- Deep-water horizontal velocity amplitude at the free surface:

$$U_m = \omega a = \frac{2\pi}{T} \frac{H}{2}$$

Step-by-step solution

1. Convert wave height to wave amplitude:

$$a = \frac{H}{2} = \frac{2}{2} = 1.0 \text{ m.}$$

2. Compute angular frequency:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{8} = 0.785 \text{ rad/s.}$$

3. Compute the maximum horizontal particle velocity at the surface:

$$U_m = \omega a = (0.785)(1.0) = 0.785 \text{ m/s.}$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

W4. Maximum wave height before breaking

FLOW AROUND AND FORCES ON CYLINDERS. What is the maximum wave height on a horizontal bed given a wave period $T = 7$ s and a water depth $h = 19$ m?

Options:

- A. $H_{\max} = 15.2$ m
- B. $H_{\max} = 9.3$ m
- C. $H_{\max} = 4.1$ m

Correct answer: B. $H_{\max} = 9.3$ m

Formulario / principles used

- Angular frequency:

$$\omega = \frac{2\pi}{T}$$

- Linear wave dispersion relation:

$$\omega^2 = gk \tanh(kh)$$

- Wavelength:

$$L = \frac{2\pi}{k}$$

- Miche breaking criterion:

$$H_{\max} = 0.142 L \tanh(kh)$$

Step-by-step solution

1. Calculate angular frequency:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{7} = 0.898 \text{ rad/s.}$$

2. Solve the dispersion relation numerically:

$$\omega^2 = gk \tanh(kh)$$

$$(0.898)^2 = 9.81 k \tanh(19k).$$

3. This gives approximately:

$$k \approx 0.0847 \text{ m}^{-1}.$$

4. Calculate the wavelength:

$$L = \frac{2\pi}{k} = \frac{2\pi}{0.0847} = 74.2 \text{ m.}$$

5. Calculate
- kh
- and
- $\tanh(kh)$
- :

$$kh = (0.0847)(19) = 1.61,$$

$$\tanh(kh) = \tanh(1.61) \approx 0.92.$$

6. Apply Miche criterion:

$$H_{\max} = 0.142(74.2)(0.92) = 9.69 \text{ m.}$$

7. The closest available answer is:

$$H_{\max} = 9.3 \text{ m.}$$

Source(s): *Test_Exam_all / Block 2 reference PDF*

W5. True wave period from encountered period

WAVES: A ship sails with a speed of 18 knots. The ship encounters regular waves at an angle $\beta = 135^\circ$ (bow-quartering waves). On the ship, the encountered wave period is $T_{enc} = 10$ s. What is the absolute (true) wave period T , as measured by a fixed observer relative to the waves?

Options:

A. $T = 12.5$ s

B. $T = 13.0$ s

C. $T = 13.5$ s

Correct answer: B. $T = 13.0$ s

Formulario / principles used

- Encounter frequency in deep water:

$$\omega_e = \omega - \frac{\omega^2}{g} U \cos \beta$$

- Encounter period:

$$\omega_e = \frac{2\pi}{T_{enc}}$$

- True wave period:

$$T = \frac{2\pi}{\omega}$$

- Speed conversion:

$$U \text{ [m/s]} = 0.51444 U \text{ [knots]}$$

Step-by-step solution

1. Convert the ship speed:

$$U = 18(0.51444) = 9.26 \text{ m/s.}$$

2. Convert encountered period to encountered angular frequency:

$$\omega_e = \frac{2\pi}{T_{enc}} = \frac{2\pi}{10} = 0.628 \text{ rad/s.}$$

3. Compute the heading cosine:

$$\cos(135^\circ) = -0.707.$$

4. Insert into the encounter-frequency relation:

$$0.628 = \omega - \frac{\omega^2}{9.81}(9.26)(-0.707).$$

5. Since the cosine is negative, the second term becomes positive:

$$0.628 = \omega + 0.667\omega^2.$$

6. Rearrange as a quadratic:

$$0.667\omega^2 + \omega - 0.628 = 0.$$

7. Solve for the positive root:

$$\omega = \frac{-1 + \sqrt{1 + 4(0.667)(0.628)}}{2(0.667)} = 0.477 \text{ rad/s.}$$

8. Convert to true period:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{0.477} = 13.18 \text{ s.}$$

9. The closest option is:

$$T = 13.0 \text{ s.}$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

W6. Encountered period in head seas

WAVE THEORY - Encountered Period (Head Seas). A ship travels at 20 knots in head seas ($\beta = 180^\circ$). The waves have frequency $f = 0.1$ Hz. What is the encountered period T_e ?

Options:

A. $T_e = 8.2$ s

B. $T_e = 6.9$ s

C. $T_e = 6.0$ s

Correct answer: C. $T_e = 6.0$ s

Formulario / principles used

- Wave angular frequency:

$$\omega = 2\pi f$$

- Encounter angular frequency:

$$\omega_e = \omega - \frac{\omega^2}{g} U \cos \beta$$

- Head seas:

$$\beta = 180^\circ, \quad \cos(180^\circ) = -1$$

- Encountered period:

$$T_e = \frac{2\pi}{\omega_e}$$

Step-by-step solution

- Convert the ship speed:

$$U = 20(0.51444) = 10.29 \text{ m/s.}$$

- Compute wave angular frequency:

$$\omega = 2\pi f = 2\pi(0.1) = 0.628 \text{ rad/s.}$$

- In head seas, $\cos \beta = -1$, so:

$$\omega_e = \omega + \frac{\omega^2}{g} U.$$

- Substitute numerically:

$$\omega_e = 0.628 + \frac{(0.628)^2}{9.81}(10.29).$$

- Compute the added encounter-frequency term:

$$\frac{(0.628)^2}{9.81}(10.29) = \frac{0.394}{9.81}(10.29) = 0.414 \text{ rad/s.}$$

- Therefore:

$$\omega_e = 0.628 + 0.414 = 1.042 \text{ rad/s.}$$

- Convert to encountered period:

$$T_e = \frac{2\pi}{1.042} = 6.03 \text{ s} \approx 6.0 \text{ s.}$$

Wave statistics and extreme values

S1. Significant wave height from a wave record

WAVES: From a wave record consisting of 141 waves (trough to crest), the number of waves of various heights are as reported in the table below. What is the significant wave height?

H [m]	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
N	11	38	45	27	12	4	3	1

Options:

A. $H_s = 2.25$ m

B. $H_s = 2.15$ m

C. $H_s = 2.35$ m

Correct answer: C. $H_s = 2.35$ m

Formulario / principles used

- Significant wave height from a wave record:

$$H_s = H_{1/3} = \text{mean height of the highest one-third of all waves}$$

- Number of waves in the top third:

$$N_{1/3} = \frac{N_{\text{total}}}{3}$$

Step-by-step solution

- Compute the number of waves in the highest one-third:

$$N_{1/3} = \frac{141}{3} = 47 \text{ waves.}$$

- Select the 47 highest waves, starting from the largest wave-height class:

$$4.0 \text{ m} : 1, \quad 3.5 \text{ m} : 3, \quad 3.0 \text{ m} : 4, \quad 2.5 \text{ m} : 12.$$

- Count these waves:

$$1 + 3 + 4 + 12 = 20 \text{ waves.}$$

- More waves needed:

$$47 - 20 = 27 \text{ waves.}$$

- The next class is 2.0 m, which contains exactly 27 waves.

- Compute the mean height of the selected 47 waves:

$$H_s = \frac{1(4.0) + 3(3.5) + 4(3.0) + 12(2.5) + 27(2.0)}{47}.$$

- Substitute and add:

$$H_s = \frac{4.0 + 10.5 + 12.0 + 30.0 + 54.0}{47} = \frac{110.5}{47} = 2.35 \text{ m.}$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

S2. Significant wave height from a sketched spectrum

WAVE STATISTICS - Significant Wave Height from Spectrum. A wave spectrum $S(\omega)$ has been sketched. The peak spectral density is $S_{\max} \approx 5 \text{ m}^2\text{s}/\text{rad}$ and the area under the curve is approximately $m_0 \approx 1 \text{ m}^2$. Estimate H_s .

Options:

A. $H_s = 2 \text{ m}$

B. $H_s = 6 \text{ m}$

C. $H_s = 4 \text{ m}$

Correct answer: C. $H_s = 4 \text{ m}$

Formulario / principles used

- Zeroth spectral moment:

$$m_0 = \int_0^\infty S(\omega) d\omega$$

- Significant wave height from spectral moment:

$$H_s = 4\sqrt{m_0}$$

Step-by-step solution

1. Interpret the area under the wave spectrum as the zeroth spectral moment:

$$m_0 = \int_0^\infty S(\omega) d\omega.$$

2. The question states that the area is approximately:

$$m_0 \approx 1 \text{ m}^2.$$

3. Use the significant-wave-height relation:

$$H_s = 4\sqrt{m_0}.$$

4. Substitute numerically:

$$H_s = 4\sqrt{1} = 4 \text{ m}.$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

S3. Crest exceedance from Rayleigh distribution

WAVE STATISTICS. For an irregular wave record, the variance of the surface elevation is $m_0 = 1.44 \text{ m}^2$. What is the probability that a wave crest exceeds the level $h_1 = 1.2 \text{ m}$?

Options:

A. 13.5%

B. 36.8%

C. 60.7%

D. 77.9%

Correct answer: C. 60.7%

Formulario / principles used For Rayleigh-distributed crest amplitudes:

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2m_0}\right)$$

where m_0 is the variance of the surface elevation and h_1 is the crest level threshold.

Step-by-step solution

1. Square the threshold level:

$$h_1^2 = (1.2)^2 = 1.44 \text{ m}^2.$$

2. Compute the denominator:

$$2m_0 = 2(1.44) = 2.88 \text{ m}^2.$$

3. Substitute in the Rayleigh exceedance formula:

$$P(Z > h_1) = \exp\left(-\frac{1.44}{2.88}\right) = \exp(-0.5).$$

4. Compute the probability:

$$P(Z > h_1) = 0.607 = 60.7\%.$$

Source(s): 41107_mock_exam_with_answers / Block 2 reference PDF

S4. Gaussian exceedance from acceleration variance

WAVE STATISTICS - Rayleigh Distribution & Exceedance Probability. An acceleration spectrum has variance $\sigma^2 = m_0 = 1.5 \text{ (m/s}^2\text{)}^2$. What is the probability of exceeding an acceleration level $a = 2.5 \text{ m/s}^2$?

Options:

A. $p_a = 5\%$

B. $p_a = 12\%$

C. $p_a \approx 2\%$

Correct answer: C. $p_a \approx 2\%$

Important distinction. The source question labels Rayleigh, but the selected answer follows a Gaussian process-value exceedance using a standard normal table. A Rayleigh peak exceedance with the same numbers would give approximately 12.5%.

Formulario / principles used

- Standard deviation from variance:

$$\sigma = \sqrt{m_0}$$

- Standard normal variable:

$$z = \frac{a}{\sigma}$$

- Gaussian exceedance probability:

$$P(A > a) = P(Z > z)$$

Step-by-step solution

1. Compute the standard deviation:

$$\sigma = \sqrt{m_0} = \sqrt{1.5} = 1.225 \text{ m/s}^2.$$

2. Compute the standard normal value:

$$z = \frac{a}{\sigma} = \frac{2.5}{1.225} = 2.04.$$

3. From a standard normal table:

$$P(Z > 2.04) \approx 0.021.$$

4. Convert to percent:

$$0.021 = 2.1\% \approx 2\%.$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

S5. Most probable largest wave height

SPECTRAL ANALYSIS - Most Probable Largest Wave Height. A 3-hour storm sea state has $H_s = 4$ m and mean zero-crossing period $T_z = 8$ s. What is the most probable largest wave height $H_{\max}$ in this storm?

Options:

A. $H_{\max} \approx 6.3$ m

B. $H_{\max} \approx 7.4$ m

C. $H_{\max} \approx 8.5$ m

Correct answer: B. $H_{\max} \approx 7.4$ m

Formulario / principles used

- Number of waves during the storm:

$$N = \frac{T_{\text{storm}}}{T_z}$$

- Most probable largest wave height:

$$H_{\text{MPL}} = \frac{H_s}{2} \sqrt{2 \ln N}$$

Step-by-step solution

1. Convert storm duration to seconds:

$$T_{\text{storm}} = 3 \text{ h} = 3(3600) = 10800 \text{ s.}$$

2. Compute the number of waves:

$$N = \frac{10800}{8} = 1350.$$

3. Compute the logarithmic factor:

$$\ln N = \ln(1350) = 7.21.$$

4. Compute the square-root term:

$$\sqrt{2 \ln N} = \sqrt{2(7.21)} = \sqrt{14.42} = 3.80.$$

5. Compute the most probable largest wave height:

$$H_{\text{MPL}} = \frac{H_s}{2} \sqrt{2 \ln N} = \frac{4}{2}(3.80) = 2(3.80) = 7.6 \text{ m.}$$

6. The closest available option is:

$$H_{\text{max}} \approx 7.4 \text{ m.}$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

Spectral moments and spectral response

M1. Representative period from H_s and first spectral moment

SPECTRAL ANALYSIS - Spectral Moments and Wave Period. A wave system has $H_s = 3.5$ m and first spectral moment $m_1 = 0.5$ m² rad/s. What is a representative wave period T ?

Options:

- A. $T = 9.6$ s
- B. $T = 6.8$ s
- C. $T = 14.1$ s

Correct answer: A. $T = 9.6$ s

Formulario / principles used

- Spectral moments:

$$m_n = \int_0^\infty \omega^n S(\omega) d\omega$$

- Significant wave height relation:

$$H_s = 4\sqrt{m_0} \Rightarrow m_0 = \left(\frac{H_s}{4}\right)^2$$

- First mean period:

$$T_1 = 2\pi \frac{m_0}{m_1}$$

Step-by-step solution

1. Compute m_0 from significant wave height:

$$m_0 = \left(\frac{H_s}{4}\right)^2 = \left(\frac{3.5}{4}\right)^2 = (0.875)^2 = 0.766 \text{ m}^2.$$

2. Use the first mean period formula:

$$T_1 = 2\pi \frac{m_0}{m_1}.$$

3. Substitute numerically:

$$T_1 = 2\pi \frac{0.766}{0.5} = \frac{4.81}{0.5} = 9.62 \text{ s}.$$

4. The closest option is:

$$T = 9.6 \text{ s}.$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

M2. Representative mean period variant

WAVE STATISTICS / SPECTRAL MOMENTS. A wave system has $H_s = 4.0$ m and first spectral moment $m_1 = 0.80$ m² rad/s. What is a representative mean period T_1 approximately?

Options:

- A. 6.3 s

- B. 7.9 s
C. 9.4 s
D. 12.6 s

Correct answer: B. 7.9 s

Formulario / principles used

$$m_0 = \left(\frac{H_s}{4} \right)^2$$

$$T_1 = 2\pi \frac{m_0}{m_1}$$

Step-by-step solution

1. Compute m_0 :

$$m_0 = \left(\frac{4.0}{4} \right)^2 = (1.0)^2 = 1.0 \text{ m}^2.$$

2. Compute T_1 :

$$T_1 = 2\pi \frac{1.0}{0.80} = \frac{6.283}{0.80} = 7.85 \text{ s.}$$

3. The closest option is:

$$T_1 \approx 7.9 \text{ s.}$$

Source(s): 41107_mock_exam_with_answers / Block 2 reference PDF

R1. Transfer function magnitude from spectral densities

SPECTRAL ANALYSIS: Information used for spectral calculations of a ship is available. Specifically, it is known that, at the frequency ω_a , the wave energy spectral density is $S(\omega_a) = 2.3 \text{ m}^2/\text{rad}$ and the corresponding pitch energy spectral density is $R(\omega_a) = 0.006 \text{ rad}^2/\text{s}/\text{rad}$. What value, approximately, has the magnitude $|\Phi(\omega_a)|$ of the pitch transfer function at the frequency ω_a ?

Options:

- A. $|\Phi(\omega_a)| = 2.9 \text{ deg/m}$
B. $|\Phi(\omega_a)| = 1.3 \text{ deg/m}$
C. $|\Phi(\omega_a)| = 0.15 \text{ deg/m}$

Correct answer: A. $|\Phi(\omega_a)| = 2.9 \text{ deg/m}$

Formulario / principles used

- Response spectrum relation:

$$R(\omega) = |\Phi(\omega)|^2 S(\omega)$$

- Transfer function magnitude:

$$|\Phi(\omega)| = \sqrt{\frac{R(\omega)}{S(\omega)}}$$

- Convert from rad/m to deg/m:

$$1 \text{ rad} = \frac{180}{\pi} \text{ deg}$$

Step-by-step solution

1. Compute the ratio:

$$\frac{R(\omega_a)}{S(\omega_a)} = \frac{0.006}{2.3} = 0.002609 \text{ rad}^2/\text{m}^2.$$

2. Take the square root:

$$|\Phi(\omega_a)| = \sqrt{0.002609} = 0.0511 \text{ rad/m}.$$

3. Convert to degrees per metre:

$$0.0511 \frac{\text{rad}}{\text{m}} \cdot \frac{180}{\pi} = 2.93 \text{ deg/m}.$$

4. The closest option is:

$$|\Phi(\omega_a)| = 2.9 \text{ deg/m}.$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

R2. Transfer function magnitude - numerical variant

SPECTRAL ANALYSIS. At a frequency ω_a , the wave energy spectral density is $S(\omega_a) = 1.8 \text{ m}^2\text{s}/\text{rad}$ and the corresponding pitch response spectral density is $R(\omega_a) = 0.008 \text{ rad}^2\text{s}/\text{rad}$. What is the magnitude of the pitch transfer function $|\Phi(\omega_a)|$ approximately?

Options:

- A. 0.0667 rad/m (3.8 deg/m)
- B. 0.0383 rad/m (2.2 deg/m)
- C. 0.0067 rad/m (0.38 deg/m)
- D. 0.1067 rad/m (6.1 deg/m)

Correct answer: A. 0.0667 rad/m (3.8 deg/m)

Formulario / principles used

$$R(\omega) = |\Phi(\omega)|^2 S(\omega)$$

$$|\Phi(\omega)| = \sqrt{\frac{R(\omega)}{S(\omega)}}$$

$$|\Phi|_{\text{deg/m}} = |\Phi|_{\text{rad/m}} \frac{180}{\pi}$$

Step-by-step solution

1. Compute the spectral-density ratio:

$$\frac{R}{S} = \frac{0.008}{1.8} = 0.004444.$$

2. Take the square root:

$$|\Phi| = \sqrt{0.004444} = 0.0667 \text{ rad/m}.$$

3. Convert to degrees per metre:

$$|\Phi| = 0.0667 \frac{180}{\pi} = 3.82 \text{ deg/m}.$$

Source(s): 41107_mock_exam_with_answers / Block 2 reference PDF

R3. Response spectral density from wave spectrum and transfer function

SPECTRAL ANALYSIS - Response PSD from Wave Spectrum. At frequency ω_a , wave spectral density $S(\omega_a) = 1.8 \text{ m}^2\text{s/rad}$ and pitch transfer function magnitude $Y(\omega_a) = 0.08 \text{ rad/m}$. What is the pitch energy spectral density $R(\omega_a)$?

Options:

- A. $R(\omega_a) = 0.01 \text{ deg}^2\text{s/rad}$
- B. $R(\omega_a) = 8 \text{ deg}^2\text{s/rad}$
- C. $R(\omega_a) = 38 \text{ deg}^2\text{s/rad}$

Correct answer: C. $R(\omega_a) = 38 \text{ deg}^2\text{s/rad}$

Formulario / principles used

- Response spectral density:

$$R(\omega) = |Y(\omega)|^2 S(\omega)$$

- Convert transfer function from radians to degrees:

$$Y_{\text{deg/m}} = Y_{\text{rad/m}} \frac{180}{\pi}$$

Step-by-step solution

1. Convert Y to degrees per metre:

$$Y_{\text{deg/m}} = 0.08 \frac{180}{\pi} = 4.584 \text{ deg/m}.$$

2. Square the transfer function magnitude:

$$Y^2 = (4.584)^2 = 21.0 \text{ deg}^2/\text{m}^2.$$

3. Multiply by the wave spectral density:

$$R(\omega_a) = Y^2 S(\omega_a) = (21.0)(1.8) = 37.8 \text{ deg}^2\text{s/rad}.$$

4. The closest option is:

$$R(\omega_a) = 38 \text{ deg}^2\text{s/rad}.$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

R4. Significant heave amplitude of a small boat

SPECTRAL ANALYSIS: A wave system is described by a Bretschneider wave spectrum with $H_s = 2$ m and $T_p = 10$ s. A small boat of length $L = 8$ m is sailing in the wave system. What is the significant amplitude h_{heave} of the boat's heave motion approximately?

Options:

A. $h_{\text{heave}} = 4$ m

B. $h_{\text{heave}} = 1$ m

C. $h_{\text{heave}} = 2$ m

Correct answer: B. $h_{\text{heave}} = 1$ m

Formulario / principles used

- Deep-water peak wavelength estimate:

$$L_p = \frac{gT_p^2}{2\pi}$$

- For a small boat relative to the wavelength:

$$|\Phi_{\text{heave}}| \approx 1$$

- Significant wave amplitude:

$$\zeta_{a,s} = \frac{H_s}{2}$$

- If $|\Phi_{\text{heave}}| \approx 1$:

$$h_{\text{heave}} \approx \zeta_{a,s} = \frac{H_s}{2}$$

Step-by-step solution

1. Estimate the peak wavelength:

$$L_p = \frac{gT_p^2}{2\pi} = \frac{9.81(10)^2}{2\pi} = \frac{981}{6.283} = 156 \text{ m.}$$

2. Compare the boat length with the peak wavelength:

$$L_{\text{boat}} = 8 \text{ m} \ll L_p = 156 \text{ m.}$$

3. Therefore the small boat approximately follows the wave surface in heave:

$$|\Phi_{\text{heave}}| \approx 1.$$

4. Significant wave amplitude is half the significant wave height:

$$\zeta_{a,s} = \frac{H_s}{2} = \frac{2}{2} = 1 \text{ m.}$$

5. Hence:

$$h_{\text{heave}} \approx 1 \text{ m.}$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

Final formula reminder for this group

Regular wave theory

$$\omega = \frac{2\pi}{T} = 2\pi f, \quad k = \frac{2\pi}{L}, \quad c = \frac{L}{T} = \frac{\omega}{k}$$

$$\omega^2 = gk \tanh(kh)$$

$$L_0 = \frac{gT^2}{2\pi}, \quad c = \frac{gT}{2\pi}, \quad c_g = \frac{c}{2} \quad (\text{deep water})$$

$$a = \frac{H}{2}, \quad U_m = \omega a = \frac{2\pi}{T} \frac{H}{2}$$

$$H_{\max} = 0.142 L \tanh(kh)$$

Encounter frequency

$$\omega_e = \omega - \frac{\omega^2}{g} U \cos \beta$$

$$T_e = \frac{2\pi}{\omega_e}, \quad T = \frac{2\pi}{\omega}$$

$$U \text{ [m/s]} = 0.51444 U \text{ [knots]}$$

Wave statistics

$H_s = H_{1/3}$ = mean height of the highest one-third waves

$$m_0 = \int_0^\infty S(\omega) d\omega, \quad H_s = 4\sqrt{m_0}$$

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2m_0}\right)$$

$$H_{\text{MPL}} = \frac{H_s}{2} \sqrt{2 \ln N}, \quad N = \frac{T_{\text{storm}}}{T_z}$$

Spectral moments and response spectra

$$m_n = \int_0^\infty \omega^n S(\omega) d\omega$$

$$m_0 = \left(\frac{H_s}{4}\right)^2, \quad T_1 = 2\pi \frac{m_0}{m_1}$$

$$R(\omega) = |\Phi(\omega)|^2 S(\omega)$$

$$|\Phi(\omega)| = \sqrt{\frac{R(\omega)}{S(\omega)}}$$

$$1 \text{ rad} = \frac{180}{\pi} \text{ deg}$$